

Assignment V

AI Compute Platforms, Applications and Trends

Submission Include:

Student Name:

Student ID:

Assignment Title:

Question I:

Scenario: Student Information Web Application

1. Create a simple web page called **Student Portal**.

Version 1 Page must display:

Student Portal

Version 1

Welcome to the Student Information System

2. Create a Docker image for the application and use an appropriate web server such as Nginx or Apache. Tag the image as:

<dockerhub-username>/student-portal:v1

3. Push Version 1 to your Docker Hub repository.

4. Then deploy the application to Kubernetes with the following requirements:

Deployment Name : student-portal

Replicas : 2

Container Port : 80

Service Type : NodePort

5. Verify that both Pods are running and access the application from a browser.

Version 2 Update

6. Modify the application so that the web page displays:

Student Portal

Version 2

Welcome to the Updated Student Information System

New Feature: Online Course Registration

7. Build a **new Docker image** and tag it as:

<dockerhub-username>/student-portal:v2

8. Push Version 2 to the **same Docker Hub repository**.

9. Without deleting and recreating the existing Kubernetes Deployment, update the application from **Version 1 to Version 2**.

10. Students must demonstrate that Kubernetes performs the update successfully and verify in the browser that **Version 2** is running.

Rollback Requirement

11. After successfully deploying Version 2, inspect the Deployment rollout history.

12. Then rollback the Deployment to the previous version **without rebuilding Version 1 and without creating a new Deployment**.

13. Verify that the browser displays:

Student Portal

Version 1

Welcome to the Student Information System

again.

Submission Requirements

Students should submit **one PDF report** containing their Student ID, Name, Class/Batch, source files (index.html and Dockerfile), commands used, and screenshots showing these key results:

1. Docker images for v1 and v2

2. Docker Hub repository containing both tags
3. Kubernetes Deployment with 2 running Pods
4. Version 1 running in the browser
5. Version 2 running after the rolling update
6. Rollout history
7. Version 1 running again after rollback